

Critical Analysis: Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models

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1 Methodological Strengths

1.1 Problem framing and operationalization of black-box uncertainty quantification

A central methodological strength lies in the explicit focus on *black-box* large language models (LLMs), where only generated text is assumed accessible, while token-level probabilities (logits) and internal representations are not. This constraint matches API-served LLM deployment settings and avoids methodological over-reliance on white-box signals that are unavailable to many practitioners. The paper further separates two often conflated targets:

- **Input uncertainty:** dispersion of plausible generations for a fixed input, formalized as $U(x)$.
- **Generation confidence:** reliability of a specific sampled response s given x , formalized as $C(x, s)$.

This separation enables evaluation protocols that distinguish “hard questions” from “bad answers to otherwise easy questions,” which aligns with selective generation and rejection-based deployment requirements.

1.2 A principled reduction from high-dimensional sequence space to semantic similarity space

The paper avoids infeasible entropy computations over the full sequence space by shifting uncertainty estimation to *semantic* relationships among multiple sampled generations. Concretely, m responses are sampled for each input, and pairwise similarities are computed using lightweight external models or string-level heuristics. The similarity matrix is then transformed into graph-based statistics, which yields a coherent family of measures under a single representation:

- constructing a weighted graph whose nodes are sampled responses,
- defining uncertainty as a graph-level dispersion statistic,
- defining confidence as a node-level “centrality” statistic for response-specific scoring.

This strategy directly targets the core challenge of natural language generation (NLG): distinct surface forms can carry equivalent meaning, so reliable uncertainty measurement must operate in semantic space rather than token space.

1.3 Method families and ablations that isolate design choices

The study systematically compares multiple similarity functions and multiple graph statistics. A key methodological advantage is the explicit observation that **similarity choice can dominate performance** more than the particular graph statistic (degree, eccentricity, eigenvalue-based dispersion). This implicitly functions as an ablation over (i) semantic similarity modeling and (ii) aggregation functional form, offering interpretability about which components drive performance.

1.4 Evaluation design with complementary metrics and calibration protocol

The experimental evaluation includes two complementary ranking-based metrics:

- **AUROC** for predicting binary correctness of an individual generation.
- **AUARC** (Area Under Accuracy-Rejection Curve) for selective generation, which directly reflects the trade-off between rejection rate and retained accuracy.

AUARC is particularly suitable for deployment-like settings where a system rejects uncertain outputs. The protocol repeats experiments 10 times and uses a per-run calibration set of 1,000 questions for hyperparameter selection of uncertainty and confidence measures, reporting mean and standard deviation. This strengthens reliability claims relative to single-split reporting.

1.5 Strong baseline coverage, including white-box comparators

Despite focusing on black-box settings, the paper benchmarks against established white-box baselines where available, including Semantic Entropy and $P(\text{true})$. This provides an anchored comparison to prior UQ literature and clarifies what is lost when moving from white-box to black-box access. The inclusion of string-based lexical similarity baselines also prevents over-crediting semantic methods for gains that could come from trivial overlap measures.

1.6 Transparency and partial external validation of correctness labeling

Correctness of generations is scored using an external judge model (gpt-3.5-turbo), with a fixed correctness threshold (0.7). The study reports an additional human verification step for the judge outputs with an accuracy of approximately 0.95. While limited in scope (see limitations), this is a concrete attempt to reduce evaluation noise and supports internal validity of reported metrics.

2 Key Limitations

2.1 Evaluation validity depends on an LLM-as-judge pipeline

The correctness label is produced by gpt-3.5-turbo with a threshold-based rule. This introduces several limitations:

- **Judge bias and sensitivity:** label assignment inherits the judge model’s failure modes, including preference for certain phrasing, partial credit behaviors, and susceptibility to prompt framing.
- **Threshold arbitrariness:** the 0.7 threshold operationalizes correctness but does not reflect task-specific utility, and no sensitivity analysis is reported for alternative thresholds.
- **Metric coupling:** AUROC and AUARC measure how well uncertainty or confidence ranks samples relative to these labels, not relative to human ground truth for all samples.

The reported human verification accuracy of approximately 0.95 reduces but does not eliminate this dependence, since it validates the judge rather than replacing it.

2.2 Restricted task scope limits generalization to open-ended NLG

The evaluation uses question answering datasets (CoQA, TriviaQA, Natural Questions) because reference answers make automatic evaluation tractable. This introduces a structural limitation: free-form NLG tasks such as summarization, dialogue, and creative generation involve multiple valid outputs and require preference-based or human evaluation. The study does not provide an empirical demonstration that the same dispersion measures correlate with quality under non-QA definitions of correctness.

2.3 Black-box assumption is partially relaxed by external semantic models

Although the target LLM is treated as black-box, the approach relies on an off-the-shelf Natural Language Inference (NLI) classifier (DeBERTa-large) to compute entailment and contradiction-based similarities. This creates a dependency on:

- **NLI domain transfer:** NLI models are trained on datasets whose distribution differs from LLM generations, especially for long, noisy, or factual responses.
- **Semantic mismatch:** entailment scores are not a complete proxy for semantic equivalence of answers, particularly for underspecified, partially correct, or multi-fact responses.

As a result, the approach is black-box with respect to the target LLM but not purely text-only in the strictest sense, because it introduces another learned model that carries its own assumptions and errors.

2.4 Incomplete reporting of decoding configuration and cost

The paper states that default generation configurations are used, but the exact decoding parameters (temperature, top- p , max tokens, stop criteria) are not fully specified in the main protocol description. This limits reproducibility and interpretability because uncertainty and dispersion statistics are strongly affected by sampling behavior. In addition, the computational cost profile is not fully characterized:

- sampling m generations per input increases inference cost by a factor of m ,
- computing pairwise similarities introduces an $O(m^2)$ overhead,
- graph Laplacian eigen-decomposition introduces additional compute, even if m is kept small.

Without a standardized runtime or dollar-cost analysis, the practical feasibility for high-throughput applications remains underspecified.

2.5 Confidence calibration is acknowledged but not fully integrated into the main evaluation

The paper notes that the proposed confidence scores in raw form are not easily interpretable and can be calibrated to approximate the probability that an answer is correct, with additional evidence in an appendix. However, the main experimental conclusions emphasize ranking performance (AUROC, AUARC) rather than calibrated probability quality (such as reliability diagrams or expected calibration error). This limits conclusions about decision-making under explicit risk constraints.

2.6 Assumptions and edge cases in semantic set construction

The NumSet approach constructs semantic equivalence classes via bi-directional entailment and a transitivity assumption. The paper explicitly notes that semantic equivalence derived from NLI has no guarantee of transitivity, and that semantic similarity is continuous rather than binary. These points motivate the spectral relaxation methods, but they also highlight a limitation: equivalence-class based uncertainty estimates can be structurally unstable under small similarity perturbations.

3 Technical Bottlenecks

3.1 Semantic similarity as the dominant bottleneck

The empirical observation that similarity choice (entailment, contradiction, Jaccard) can matter more than the graph statistic indicates a fundamental bottleneck: uncertainty quantification quality is limited by the fidelity of semantic similarity estimation between generations. Three concrete constraints follow:

- **Factuality versus semantic agreement:** multiple generations can be mutually consistent yet collectively wrong, so semantic agreement alone is not a sufficient condition for correctness.
- **Underspecification:** generations can agree at a high level while differing on critical details; entailment-based similarity can under-penalize this.
- **Negation and subtle contradictions:** string-level measures miss them; NLI models detect some but remain imperfect outside standard NLI distributions.

This bottleneck limits the ceiling of dispersion-based UQ for black-box LLMs when only generation text is available.

3.2 Sampling-based estimation trades off reliability and cost

The methodology requires multiple sampled generations per input. The paper shows that even small m (such as 3) can improve selective generation performance, while larger m further improves metrics. This exposes a technical barrier: reliable UQ requires increased inference sampling, which directly increases latency and cost for API-based models. This becomes a deployment constraint for high-throughput or real-time applications.

3.3 Graph-based aggregation introduces stability and scalability constraints

Graph Laplacian methods provide a principled spectral view of semantic clustering, but they introduce additional technical challenges:

- **Stability:** spectral quantities depend on the full similarity matrix; noisy similarities can distort eigen-structure.
- **Scalability:** pairwise similarity costs scale quadratically in m ; eigen-decomposition costs scale superlinearly in m .

Although practical settings can keep m small, the pipeline still imposes non-trivial overhead relative to single-sample generation.

3.4 Confidence versus uncertainty remains entangled under sampling artifacts

Confidence is defined for an individual response, while uncertainty is an input-level property. However, both are derived from the same set of sampled responses. This creates coupling through sampling artifacts:

- a poor initial sample set can distort both confidence scores and uncertainty scores,
- decoding randomness controls diversity, which changes dispersion independently of underlying model knowledge.

This indicates a bottleneck in separating epistemic uncertainty from decoding-induced variation when only black-box text outputs are available.

4 Research Implications

4.1 Black-box UQ is practical and informative under ranking-based objectives

The empirical results demonstrate that dispersion-based measures can serve as reliable *rankers* of response quality for selective generation. A notable reported outcome is that selecting the top 50% samples by a semantic dispersion measure (Deg with entailment similarity) can improve accuracy from 62% to around 90%. This indicates that black-box UQ can materially improve decision quality even without access to token-level probabilities, supporting deployment in API-only environments.

4.2 Confidence estimation should be treated as a first-class objective in NLG

Prior NLG UQ work often centers input uncertainty. The explicit modeling of response-specific confidence addresses a deployment-critical gap: rejection decisions are often made per generated output, not only per input. This suggests that future UQ research for NLG benefits from explicitly designing confidence measures rather than retrofitting uncertainty-only estimators.

4.3 Similarity modeling is a key research axis, not a minor implementation choice

The dominance of similarity choice in performance implies that progress in black-box UQ is tightly coupled to advances in semantic equivalence modeling for generated text. This connects UQ research to broader areas such as semantic textual similarity, entailment, factual consistency, and evaluation of generative models.

4.4 Benchmarks for UQ should move beyond correctness to utility under rejection

The use of AUARC highlights the importance of evaluating UQ under selective generation protocols, which more directly corresponds to risk-controlled deployment. This strengthens the case that future benchmarks should report rejection-accuracy trade-offs and calibrated risk coverage curves rather than single-number performance summaries.

5 Potential Research Directions

5.1 Replace general-purpose NLI similarity with task-grounded semantic agreement

A concrete direction is to design similarity functions that explicitly incorporate task constraints:

- **Answerability-aware similarity:** similarity conditioned on the question, not only between answers, to reduce false agreement among generic or irrelevant answers.

- **Factuality-aware agreement:** integrate lightweight retrieval or evidence checking so that agreement among wrong answers is penalized.
- **Domain-adapted semantic equivalence:** train or calibrate similarity models on LLM generation pairs drawn from the target domain, using targeted annotation of equivalence and contradiction.

5.2 Decouple decoding-induced diversity from uncertainty about correctness

A concrete line of work is to control or normalize for decoding randomness:

- evaluate uncertainty measures under standardized decoding regimes, including deterministic decoding and controlled stochastic decoding,
- incorporate diversity normalization, where dispersion is interpreted relative to expected diversity at the current temperature or nucleus setting.

This targets the bottleneck where dispersion reflects sampling behavior rather than knowledge uncertainty.

5.3 Cost-aware and scalable approximations for multi-sample UQ

Several engineering and algorithmic directions are immediately actionable:

- **Subquadratic similarity estimation:** approximate pairwise similarity via clustering in embedding space, locality-sensitive hashing, or sparse graph construction.
- **Streaming confidence updates:** update confidence as samples arrive and stop sampling once confidence stabilizes, enabling adaptive m per input.
- **Proxy graph statistics:** replace eigen-decomposition with cheaper surrogates that preserve ranking performance.

These address deployment feasibility for API-based models and high-throughput pipelines.

5.4 Shift from ranking-only evaluation to calibrated decision-making

The paper notes that confidence scores can be calibrated to approximate correctness probability. A concrete research direction is to integrate calibration quality into the primary evaluation:

- report reliability curves and calibration error measures alongside AUROC and AUARC,
- evaluate threshold policies under explicit risk targets, such as ensuring a minimum retained accuracy at a specified coverage.

This supports adoption in high-stakes settings where interpretability of confidence values matters.

5.5 Extend selective generation protocols beyond QA correctness

To generalize to open-ended NLG, UQ methods require task-appropriate quality definitions:

- for summarization: factual consistency and coverage,
- for dialogue: helpfulness and safety adherence,
- for long-form generation: coherence and citation support where applicable.

A concrete direction is to define selective generation targets as multi-objective utility functions and evaluate how uncertainty measures predict these utilities under rejection.

5.6 Strengthen evaluation validity via diversified judges and human protocols

A robust direction is to reduce dependence on a single LLM judge by:

- cross-judging with multiple models and measuring disagreement,
- increasing human verification coverage with stratified sampling focused on disagreement regions and high-confidence wrong cases,
- reporting judge agreement metrics and failure modes.

This produces stronger claims about the true relationship between dispersion and correctness.

6 Conclusion

This paper provides a well-motivated and methodologically coherent study of uncertainty quantification for black-box LLMs, with a notable contribution in separating input uncertainty from response confidence and operationalizing both via multi-sample semantic similarity graphs. The experimental protocol is strengthened by multiple datasets, multiple LLMs, ranking-based metrics aligned with selective generation, repeated calibration splits, and direct comparison to both black-box and white-box baselines. Empirical evidence indicates that semantic dispersion measures can meaningfully improve selective generation outcomes, including a reported increase in accuracy from 62% to around 90% when selecting the top half of samples by a dispersion-based criterion.

The most critical limitations arise from evaluation dependence on an LLM-based correctness judge, restricted task scope to QA, partial relaxation of black-box assumptions via external NLI models, and incomplete reporting of decoding configurations and cost. The dominant technical bottleneck is semantic similarity estimation, which constrains the ceiling of dispersion-based UQ when only text outputs are available. The most actionable research directions target improved task-grounded similarity modeling, cost-aware scalable multi-sample estimation, decoding-normalized uncertainty definitions, and a shift from ranking-only assessment to calibrated, risk-controlled decision-making.